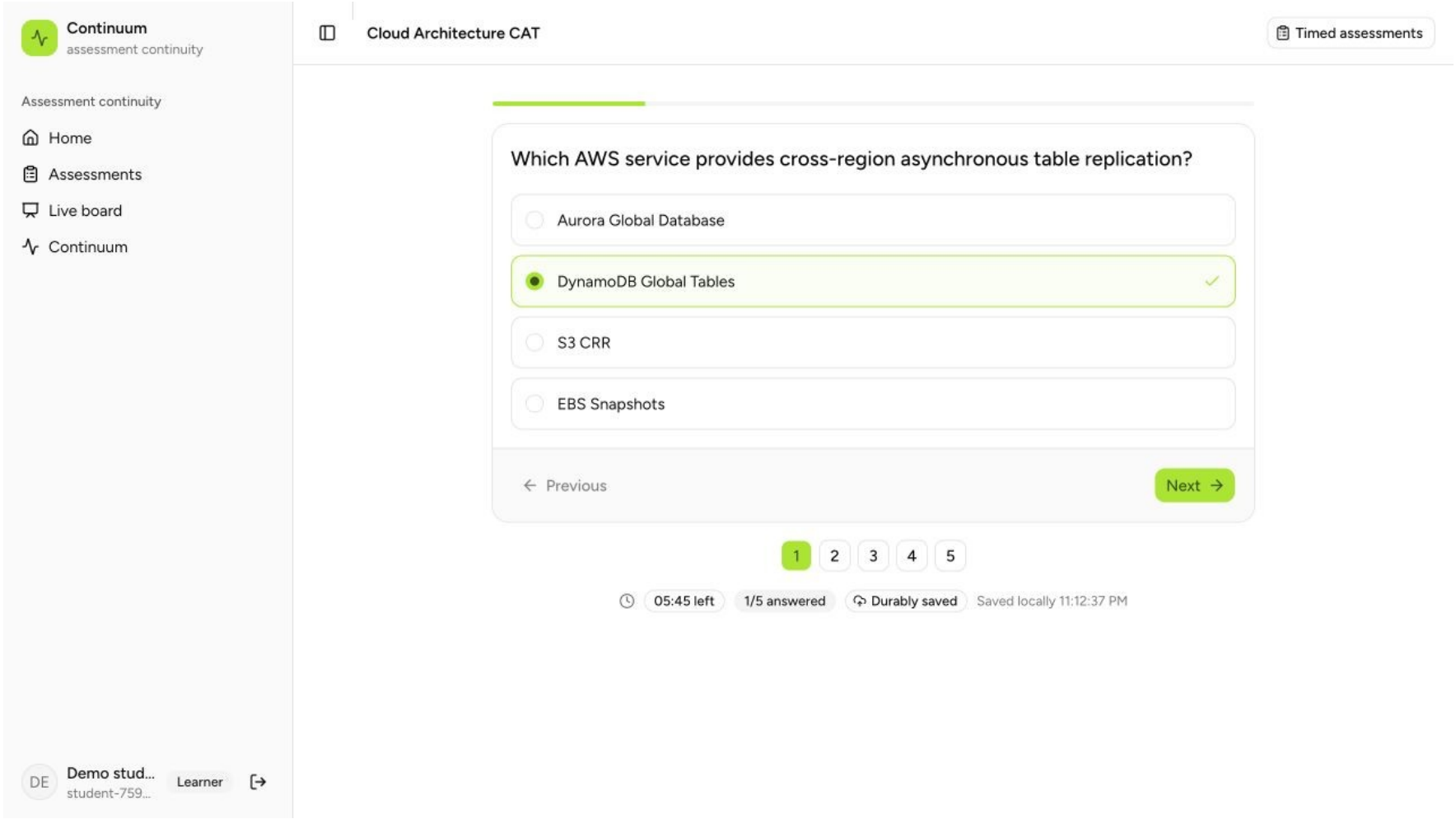
Continuum

Verified recovery for timed academic assessments

Ayush · 24BIT0394

BCSE355L · Cloud Architecture Design · Fall 2026–2027

The last minutes of an exam



Local rehearsal

15-minute server deadline

The same attempt continues through backend recovery.

Buffered answers are separate from a server receipt.

Architecture in Sydney

Layer	AWS building blocks	Continuum behavior
Student access	CloudFront / S3 + HTTP API	Exam frontend and HTTPS requests
Application route	Router Lambda + EC2 A/B	Authoritative active backend and epoch
Durable state	Regional DynamoDB	Answers, receipt, control and outbox
Recovery control	Monitor / verifier + Step Functions	Confirm, recheck, promote and reconcile

Two Availability Zones · Shared regional data and control services

Recovery policy

01 **Monitor confirmation**

Repeated deep checks distinguish a sustained fault from a blip.

02 **Locked verification**

The workflow rechecks the route and standby under an incident lease.

03 **Conditional promotion**

A newer writer epoch selects B; the public API must pass its smoke test.

04 **Safe failure handling**

Rollback requires fresh health of the former backend.

05 **Audit reconciliation**

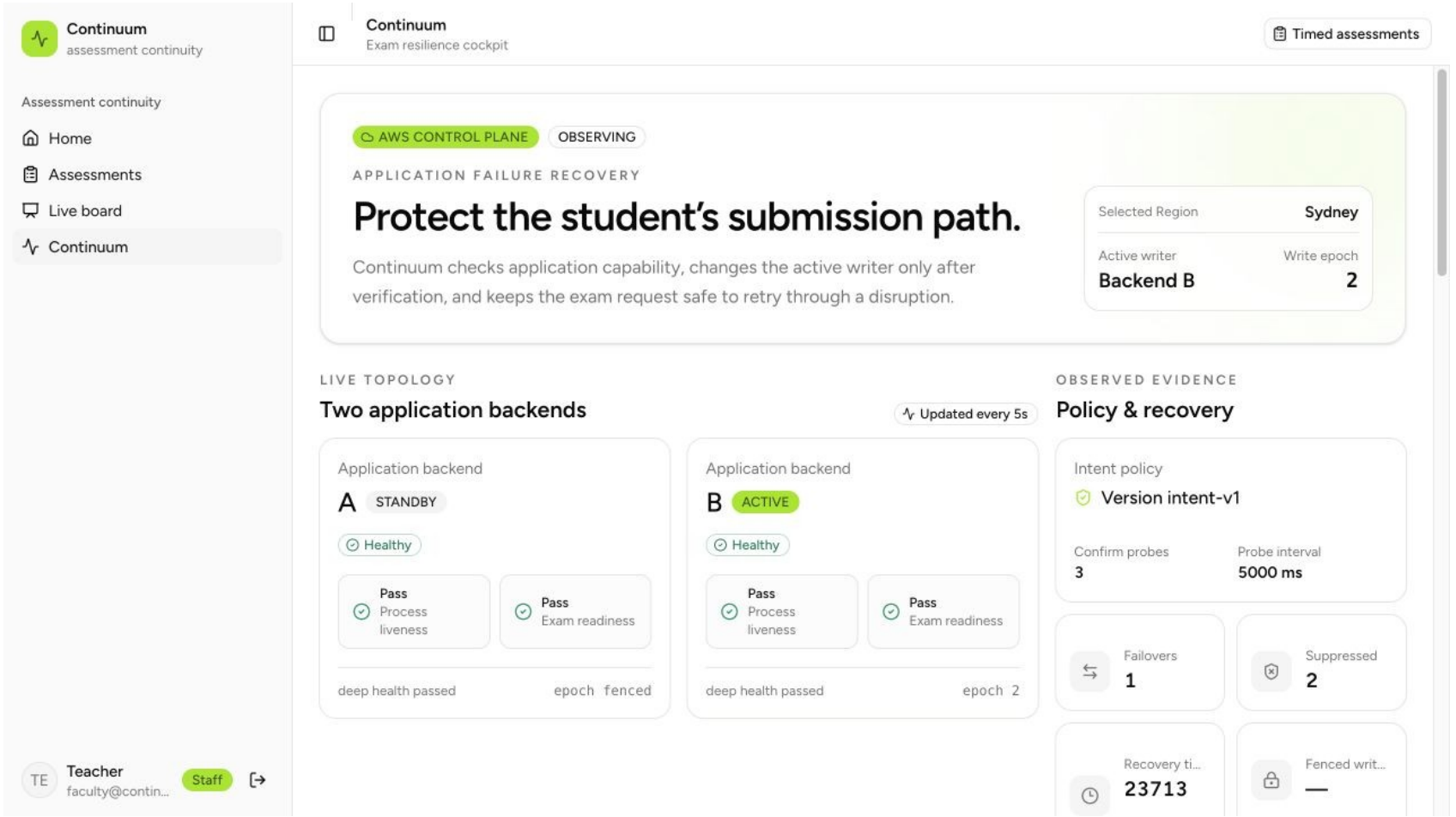
Pending committed work produces one idempotent audit summary.

A submission can outlive its response

Moment	Application contract	Observable result
Before sending	Persistent ID + immutable answer snapshot	Retries preserve the same request
Durable commit	Submission + receipt + outbox + writer check	One accepted final submission
Response lost	Exact retry returns the stored receipt	Same ID and commit time
Answers changed	Same ID with another payload is rejected	HTTP 409 conflict

Exact replay can recover a committed receipt after the exam closes.

AWS instance-stop rehearsal



50.472 s

Fault action to observed health + reconciliation

23.713 s

Monitor check cycle to public verification

One bounded Sydney rehearsal · 10 October 2026

Cost choices and trade-offs

Choice	Credit conservation	Trade-off
Two EC2 nodes	Stop between rehearsals	Retained EBS storage remains
Custom HTTP router	No always-on ALB or bought domain	Request charges, latency and limits
Regional DynamoDB	No replica promotion or cross-Region traffic	Shared regional dependency
Local rehearsal	Practice without AWS requests	One process; limited fault boundary

Free plan retained · Pausing compute does not delete storage

Live demonstration

0:00–1:00 — **Baseline**

Timer visible; A active; B ready; current epoch shown.

1:00–3:00 — **Stop the active backend**

Continue answering; observe B promotion and submit.

3:00–5:30 — **Lose a committed response**

Retry preserves the receipt reference and commit time.

5:30–7:00 — **Make the standby unsafe**

Promotion is suppressed; route and epoch stay unchanged.

7:00–8:00 — **Explain cost and scope**

Show the footprint, pause command and recovery boundary.

Evaluation and recovery boundary

Demonstrated behavior

A → B recovery after a real EC2 stop

An exact receipt after response loss

Suppressed transient / unsafe promotion

Stale writes rejected

Recovery assumptions

Regional data and control services remain available.

The original exam deadline continues.

Measured prototype results apply to this bounded application workload.

Go race checks · Frontend tests · Infrastructure validation